

BIO262 PHYSIOLOGY

Module 4

BONES

Things we forget:

- Bones make BLOOD cells – (red marrow)
- Bones store Minerals – 2/3 bone weight = Calcium Phosphate
 - Collagen is other 1/3
- CANALICULI – pathways between lacunae that allow mature osteocytes to **communicate**
- Lacunae – pockets where osteocytes reside
- Matrix – calcified calcium
- Intramembranous ossification versus endochondral....
 - Flat bone – two membranes
 - Long bone – cartilage (chondral) replacement

Flat bones (skull, sternum) = like a sandwich/Oreo cookie

❖ Made by **intramembranous ossification**

compact bone

spongy bone = diploe

compact bone

FUNCTION: Protecting internal organs

LONG bones (arms/legs, finger/toes) – like a ‘straw’;
hard bone surrounds the medullary cavity/marrow.

❖ Made by **endochondral ossification**

***cartilage at the epiphyseal plate is replaced
as bone grows....

Yellow marrow in most adult bones (adipose
tissue)

Red marrow – site of BLOOD cells formation –

FUNCTION: levers for movement, supports body

Bone Cells – Matching

1. stem cells that
differentiate into
osteoblasts

A. osteocytes

2. responsible for
osteogenesis –bone
building!

B. osteogenic/progenitor cells

C. osteoblasts

3. responsible for osteolysis
– bone
resorption/breakdown

D. osteoclasts

4. mature bone cell in the
lacunae – communicates
with other bone cells.
Controls bone remodeling
which maintains bone

**Know all of
this
information!**

Why do we need osteoclasts?

Which hormone activates these?

Why do we need osteoblasts?

Which hormone activates these?

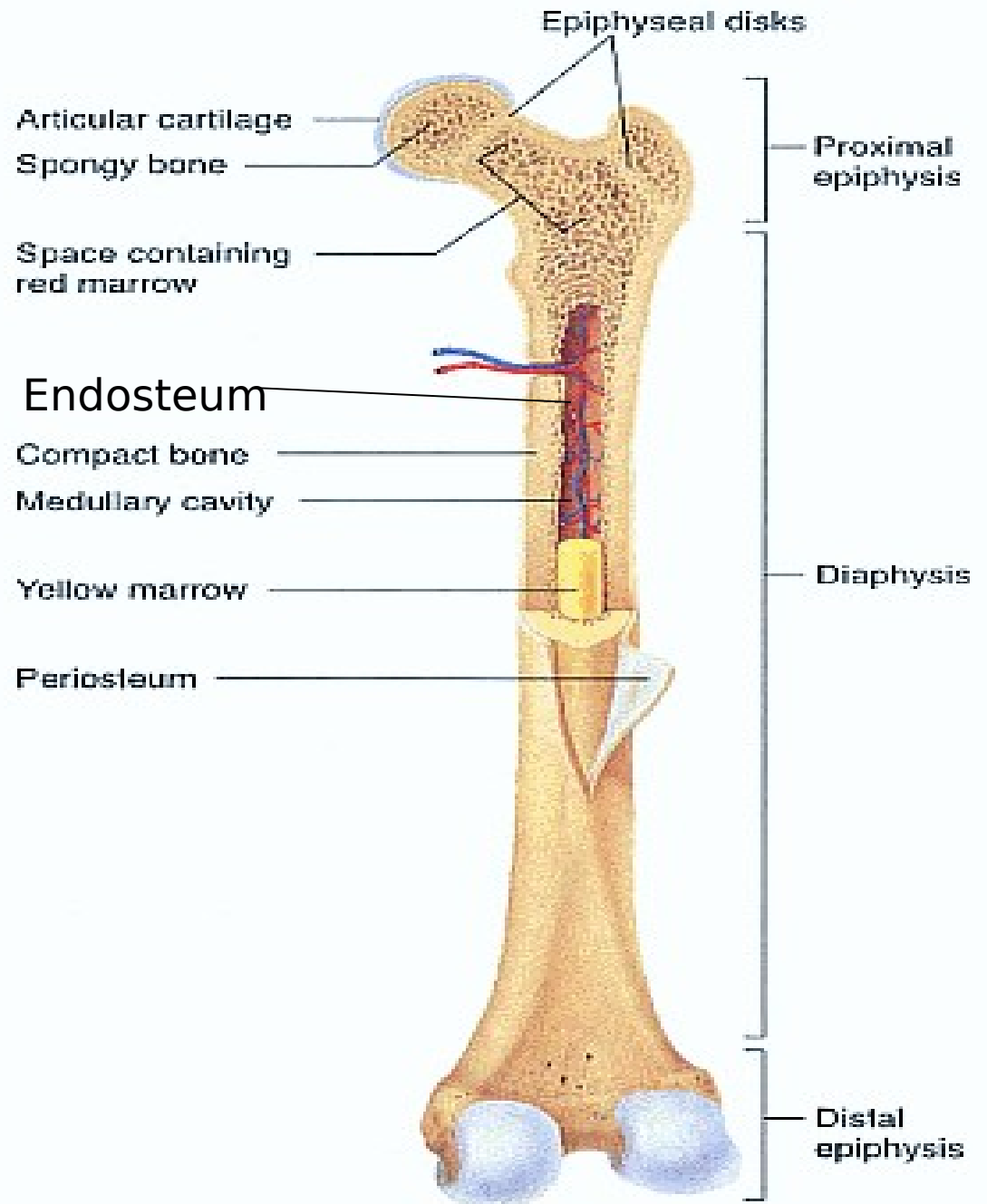
What does ossification mean? Which cell does this?

Practice Question

- You have just suffered a hairline fracture to your radius. Which two bone cells would be the most useful in the **rebuilding** process of the damaged bone? Choose all that apply
 - A. Osteocytes
 - B. Osteoblasts
 - C. Osteoclasts
 - D. Osteogenic cells

Bone Coverings

Periosteum
VS
Endosteum



Practice Question

- Holly is 5 years old. She recently went to the doctor where it was revealed that her left humerus had an epiphyseal line in place of the epiphyseal plate. Which is true?
 - A – All of Holly's bones are completely done growing.
 - B – Holly's left humerus has experienced a premature epiphyseal closure.
 - C – Intramembranous Ossification can fix this problem.
 - D – Nothing is wrong with her humerus; it is normal for there to be an epiphyseal line at age 5.



8-year-old
Notice the gaps –
this is where there
is cartilage. This
allows the bones
to elongate and
grow.
Endochondral
ossification will fill
in these gaps as
he goes through
puberty.

Endochondral Ossification vs. Intramembranous Ossification

	Endochondral	Intramembranous
Type of Bone		
Tissue Model Used		

Know this!

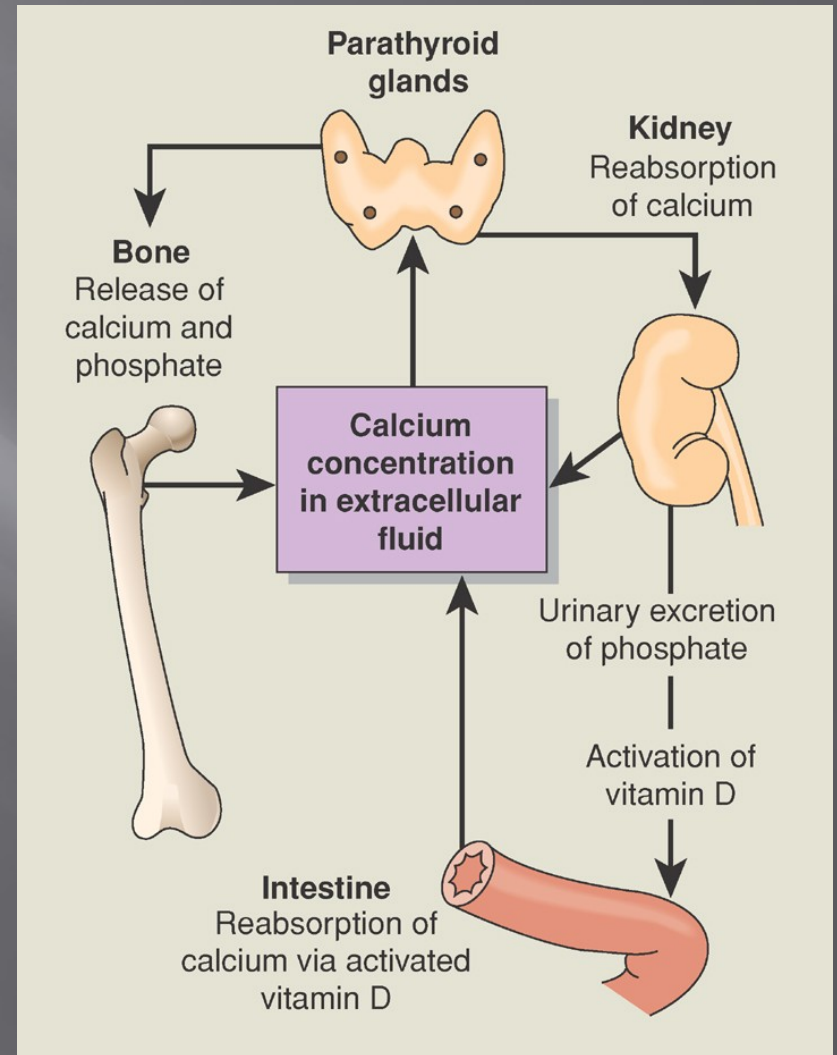
Hormones

Know all bone hormones and impact on blood Ca^{+} !

- Calcitriol (cholecalciferol + UV light = calcitriol)
- PTH
- Calcitonin

Calcium Balance

- PTH/calcitriol is released when blood calcium levels decrease
- PTH/calcitriol raises blood calcium levels three ways:
 - From bone resorption
 - From kidneys – keep it
 - From intestines - absorption



Practice Question

- Jeremy has been suffering from extremely low blood calcium levels. A test reveals a very high level of a which hormone?

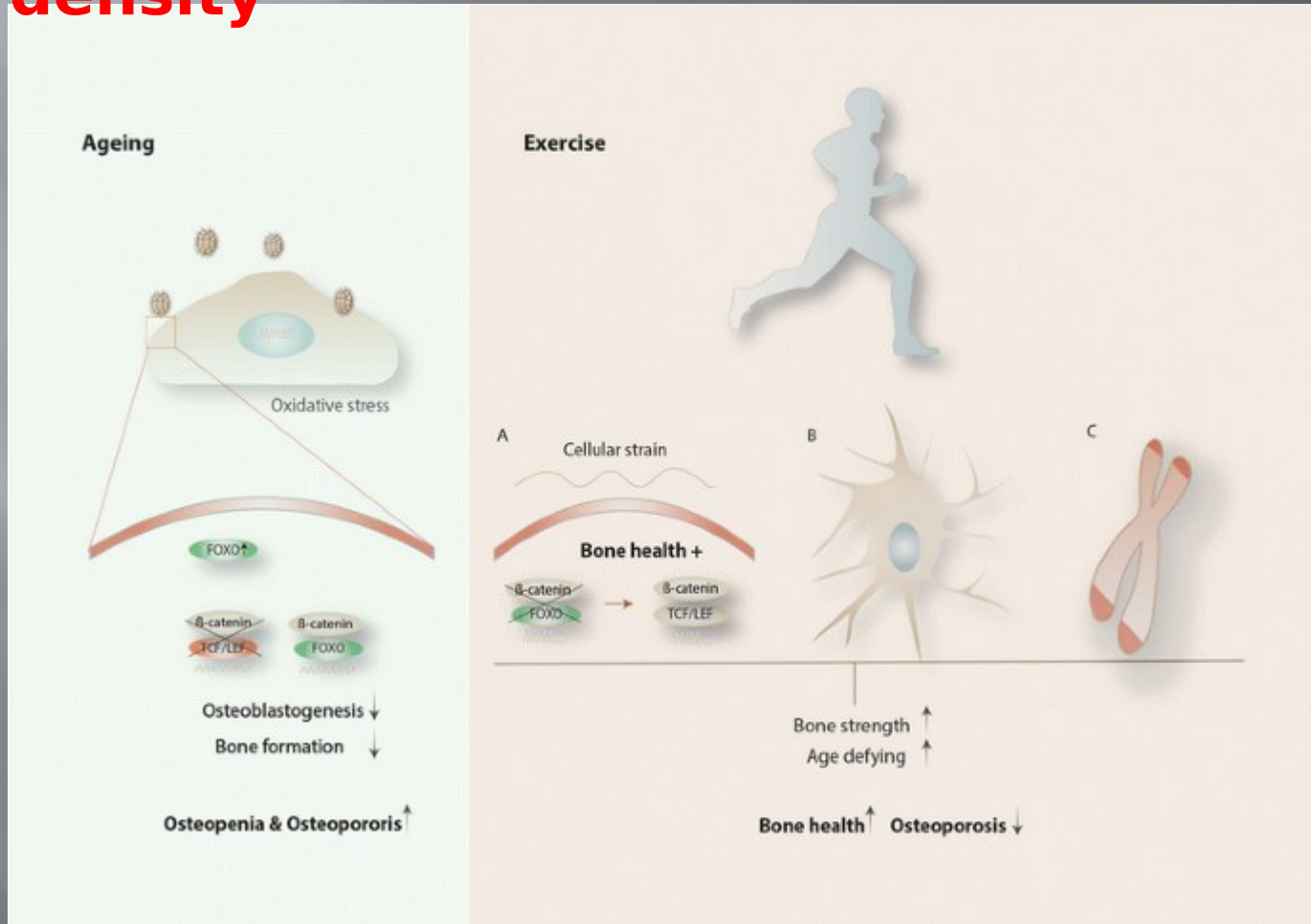
A – Calcitonin

B – Parathyroid hormone

C- Calcitriol

Impact of exercise on bone

Exercise stimulates osteoblasts which increases bone density



Reminders

DUE WEDNESDAY: 1st discussion post

DUE SUNDAY:

- Pre-lesson checkpoint
- Module 4 quiz (Respondus)
- Peer response in discussion
- Lab quiz (Respondus)
- EXAM 2 opens MONDAY – covers modules 3 & 4

You can avoid feeling overwhelmed on Sunday by completing something each day – this also leads to better retention of knowledge.